



Enhancing Thermoelectric Power Factor of PEDOT:PSS/SnSe₂ Nanocomposite Sheets by Rapid Thermal Annealing

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Abstract

Organic/inorganic nanocomposite sheets of poly (3,4-ethylenedioxythiophene)-poly (styrenesul-fonate) (PEDOT:PSS) + SnSe₂ were synthesized by a wet chemical method followed by rapid thermal annealing (RTA) at 220°C for 1 h. The ratio of PEDOT:PSS to SnSe₂ was varied by increasing incorporation of SnSe₂ powder as 0.0 g, 0.1 g, 0.2 g, and 0.3 g in the same volume of polymer solution (3 mL). It was found that the as-prepared PEDOT:PSS + SnSe₂ nanocomposite sheets were amorphous, and that the roughness of the sheets increased with increasing amounts of SnSe₂ powder. All the nanocomposite sheets showed *p*-type thermoelectric behavior. After the RTA process, the nanocomposite sheets exhibited an increased thermoelectric power factor of 37.4 $\mu\text{W m}^{-1} \text{K}^{-2}$ compared to 0.44 $\mu\text{W m}^{-1} \text{K}^{-2}$ before RTA for the same filling factor of the SnSe₂ reinforcement.

Keywords Organic–inorganic composite · thermoelectric properties · PEDOT:PSS · SnSe₂

Introduction

Organic thermoelectric (OTE) materials are recently finding increasing interest owing to their advantages of being low-cost, non-toxic, and promising flexible applicability in various fields.^{1,5} For practical high-performance application, a high dimensionless figure-of-merit (*ZT*) value is required in OTE materials. The *ZT* value is defined by;

$$ZT = \frac{S^2 T}{\rho \kappa} \quad (1)$$

where *S* is the Seebeck coefficient, ρ is the electrical resistivity, κ is the total thermal conductivity and *T* is the absolute temperature.^{6,7} The *S* and ρ values mainly control the thermionic emission (TE) power factor (PF) in^{8,9}:

$$PF = \frac{S^2}{\rho} \quad (2)$$

The poly 3, 4-ethylenedioxythiophene-polystyrene sulfonate (PEDOT-PSS) polymer is a significant area of interest in the field of OTE^{10,11} with good electrical conductivity and extremely low thermal conductivity.¹² PEDOT-PSS has been reported to have a *ZT* value of 0.42 at room temperature.¹³

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PEDOT:PSS has been intensively investigated owing to its solution process ability^{14–17} and easy tenability of its redox state (i.e., bipolar or neutral) that facilitates the optimization of the S and ρ values.^{18–22} Since incorporating a suitable reinforcement can induce remarkable enhancement in the desired properties in nanocomposites, it is imperative to investigate polymer nanocomposites of PEDOT:PSS with suitable reinforcements. In this direction, few works have been reported that improve organic-based TE material for practical use²² exhibiting the advantage of being customizable²³ by doping with several inorganic TE materials. Moreover, the annealing temperature at an optimized temperature of 220°C could enhance the PF of PEDOT:PSS films to around 162.5 times that of the pristine film before annealing within the crystalline microstructure evolution.²⁴

Among other inorganic TE materials, tin diselenide (SnSe₂) is a semiconductor material with a bulk bandgap of 1.0 eV²⁵ and which shows a mobility of 28 cm² V⁻¹ s⁻¹ in thin-film form.²⁶ SnSe₂ has additionally demonstrated a potential for applications in the fields of optoelectronics, photovoltaics, infrared optoelectronic devices, holographic recording systems, and memory-switching devices.²⁷ Moreover, the SnSe₂ material has been promoted for new 2-dimensional TE materials^{29–31} based on structure regulation^{28,29} and nano-particles,³⁰ etc. In the same way, the PEDOT:PSS has been reported to reduce the low-dimensional material

with annealing temperature.²⁴ Hence, the low-dimensional material coupling between PEDOT:PSS and SnSe₂ will be investigated in this work.

We prepared the organic/inorganic composite sheets (PEDOT:PSS + SnSe₂) through the hot plate and stirred methods based on the solution. The PEDOT:PSS + SnSe₂ composite sheets were annealed with rapid thermal annealing (RTA) at 220°C for 1 h. The microstructure, morphology, composition, and TE properties of PEDOT:PSS + SnSe₂ composite sheets investigated and reported here.

Experimental

PEDOT:PSS (Ossila PH1000) 1.0–1.3 wt% in water was used at 3 mL each to be mixed with several SnSe₂ powders (99.9%, APS < 100 nm) with 0.0 g, 0.1 g, 0.2 g, and 0.3 g. The mixing solutions were each converted into microsheets onto beakers via a hot plate at 40°C and stirred for 1 h. Then, the PEDOT:PSS + SnSe₂ polymer sheets onto beakers were stripped to obtain the sheet samples of PEDOT:PSS, PEDOT:PSS + SnSe₂ (0.1 g), PEDOT:PSS + SnSe₂ (0.2 g), and PEDOT:PSS + SnSe₂ (0.3 g). The sheet sample preparation process diagram is shown in Fig. 1. All the samples were annealed by an RTA process (Mila5000; Advanced Riko) under a vacuum state. The temperature was fixed at

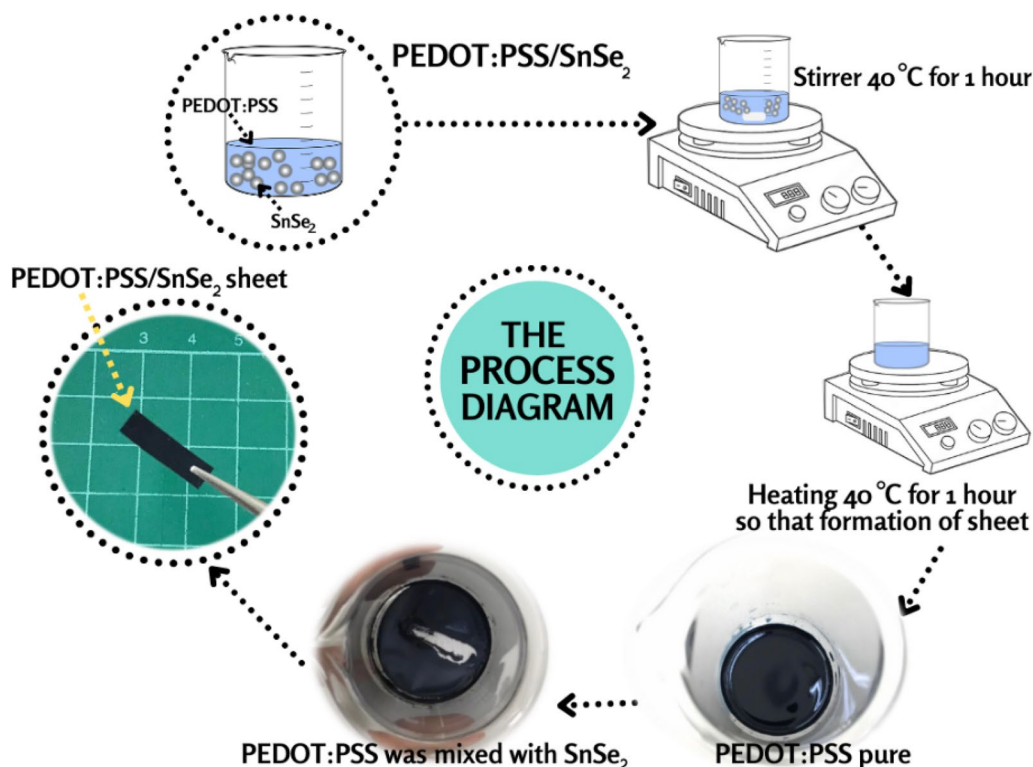


Fig. 1 Schematic of PEDOT:PSS/SnSe₂-based sheet preparation.

220°C with a heating step of 5°C/s for all the samples. After the RTA process, the microstructure of the SnSe₂ powder and polymer sheet samples was measured by x-ray diffractometer (XRD; 6100, Shimadzu) with CuK α 1 radiation in a 2 θ range of 10–70°, with scan steps of 0.02°/1 s. The morphology, thickness, and atomic composition of all the sheet samples were evaluated by the field-emission scanning electron microscopy and energy-dispersive x-ray spectroscopy (FE-SEM/EDX; SU8030; Hitachi). The carrier concentration (n) and mobility (μ) were measured by the Hall effect measurement (HMS-3000, Ecopia) based on the Van der Pauw method under a magnetic field of 0.55 T. The field ρ and S of the annealed thin-film samples were measured by a ZEM-3 apparatus (ZEM-3; Advance Riko) under a helium atmosphere at room temperature to 100°C. The sheet sample size of 3.0 $\times$ 15.0 mm² was cut and loaded onto the ZEM-3 thin film attachment.^{7,31,32}

Results and Discussion

The XRD patterns of the SnSe₂ powder and PEDOT:PSS + SnSe₂ polymer sheet samples are shown in Fig. 2. Figure 2a shows the XRD pattern of SnSe₂ powder

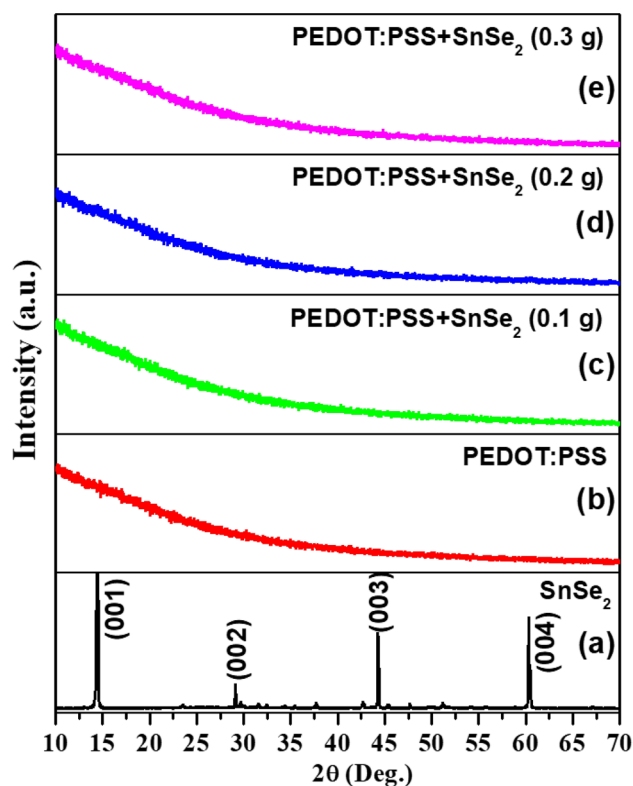


Fig. 2 XRD patterns of (a) SnSe₂ powder and (b–e) PEDOT:PSS sheets mixed with SnSe₂ powder of 0.0 g, 0.1 g, 0.2 g, and 0.3 g, respectively, before annealing.

which confirmed the single-crystal structure in hexagonal as aligned along the (001) plane (JCPDS#023-0602).³³ Figure 2b–e shows the XRD patterns of PEDOT:PSS + SnSe₂ within the SnSe₂ with added (b) 0.0 g, (c) 0.1 g, (d) 0.2 g, and (e) 0.3 g, which all indicate the amorphous phase of the polymers. Figure 3 shows the XRD pattern of the PEDOT:PSS + SnSe₂ annealed polymer sheet samples for (b) 0.0 g, (c) 0.1 g, (d) 0.2 g, and (e) 0.3 g. The effect of RTA can be seen in the composite phase of SnSe₂ with the major diffraction peaks of SnSe₂ (JCPDS#023-0602) appearing at 2 θ = 14.42°, 29.11°, 44.22°, and 60.24° corresponding to the (001), (002), (003), and (004) planes, respectively.

The morphology and chemical composition of the PEDOT:PSS + SnSe₂ polymer sheets are shown in Fig. 4. Figure 4(1) shows SEM images of the amorphous surface morphology before annealing for PEDOT:PSS within the SnSe₂ with added (a) 0.0 g, (b) 0.1 g, (c) 0.2 g, and (d) 0.3 g. As a result, the surface morphology of the PEDOT:PSS polymer sheets was more homogenized and had very smooth roughness, and then the rough surface of the PEDOT:PSS + SnSe₂ polymer sheets was increased with

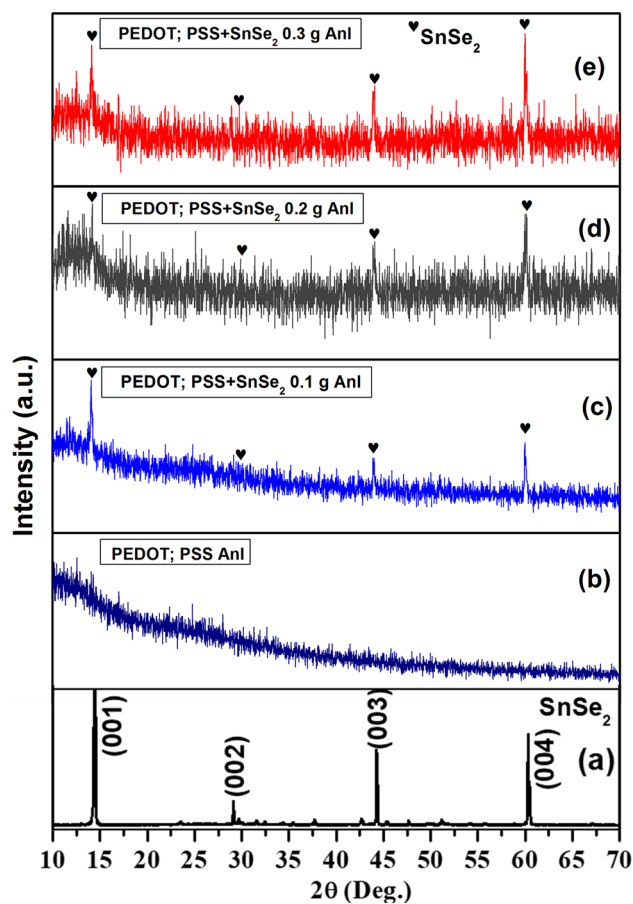


Fig. 3 XRD patterns of (a) SnSe₂ powder and (b–e) PEDOT:PSS sheets mixed with SnSe₂ powder of 0.0 g, 0.1 g, 0.2 g, and 0.3 g, respectively, after annealing.

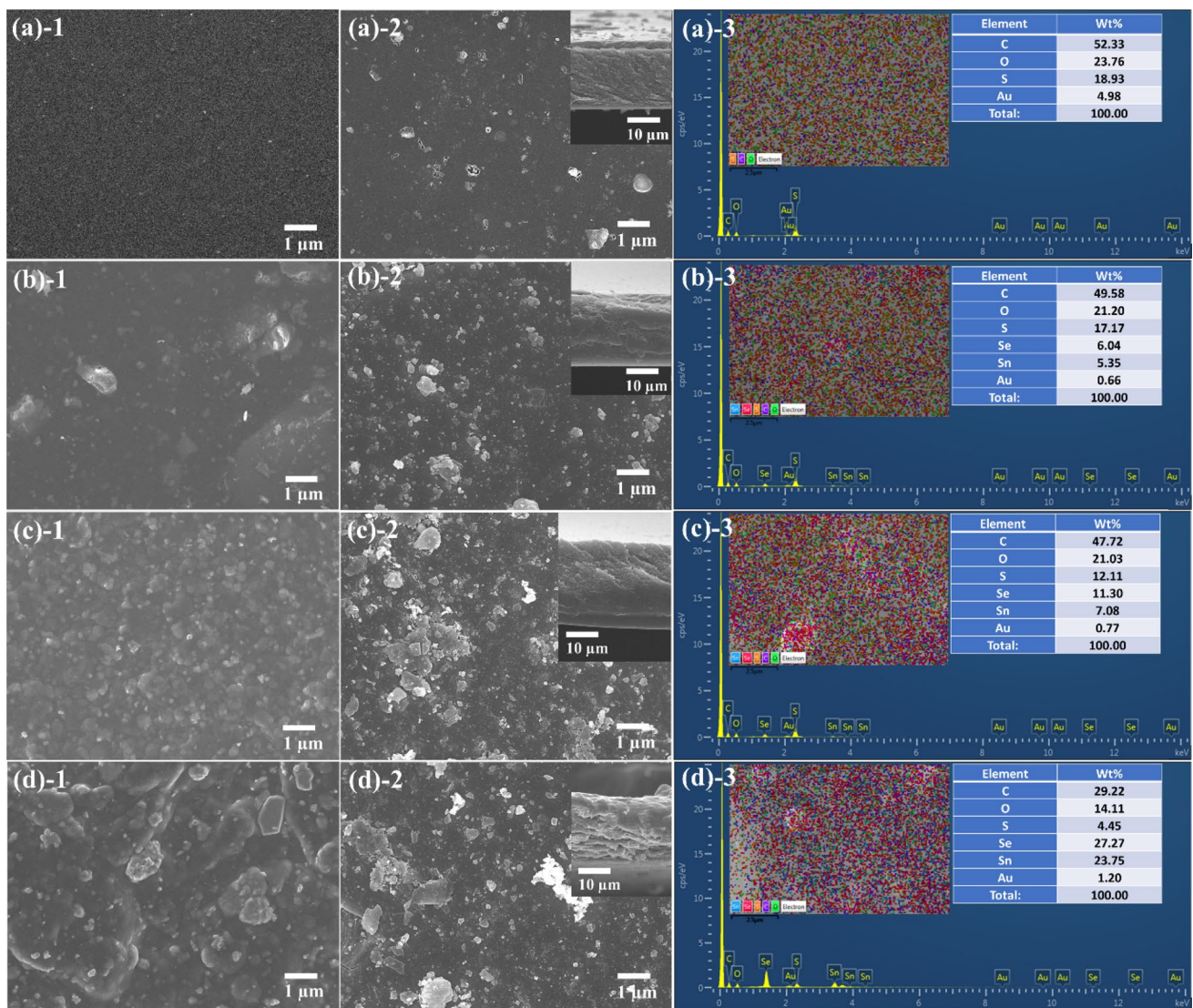


Fig. 4 SEM images of surface morphology before annealing (1), surface morphology after annealing and thickness (2) and EDX spectra with (insets) atomic mapping and composition (3) of PEDOT:PSS sheets mixed with SnSe₂ powder of (a) 0.0 g, (b) 0.1 g, (c) 0.2 g and (d) 0.3 g.

the concentration of added SnSe₂. Figure 4(2) shows SEM images of the composite surface morphology after annealing and cross-sections to obtain sheet thickness (insets) for PEDOT:PSS within the SnSe₂ with added (a) 0.0 g, (b) 0.1 g, (c) 0.2 g, and (d) 0.3 g. The results show the surface morphology of the SnSe₂ as mixed onto the PEDOT:PSS polymer sheet. These results agree with the XRD results of the composite phase of SnSe₂ as affected by the RTA process.

While all the sheet thicknesses are the same at around 30 μm , Fig. 4(3) shows EDX spectra and element mapping of the atomic composition in PEDOT:PSS + SnSe₂ within the SnSe₂ with added (a) 0.0 g, (b) 0.1 g, (c) 0.2 g, and (d) 0.3 g. The results indicate that the atomic composition in PEDOT:PSS + SnSe₂ polymer sheets is composed

of C, O, and S elements for PEDOT:PSS essential and C, O, S, Sn, and Se elements for PEDOT:PSS + SnSe₂. Then, the Au atomic composition was obtained from noble metal coating to especially helpful samples conducted during the SEM processing. Moreover, the atomic composition of Sn and Se has been increased with the increase in the concentration of SnSe₂ to confirm the composition of the PEDOT:PSS + SnSe₂ polymer sheets.

Figure 5 shows the n and μ values measured by the Hall effect method at room temperature of the PEDOT:PSS + SnSe₂ polymer sheets, indicating that the samples before annealing had very low mobility (0.014–0.15 $\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$). Then, after the annealing of the samples, the μ has reached a maximum of 1.84 $\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$ (PEDOT:PSS + SnSe₂ (0.2 g)). Hence, the reaching of μ

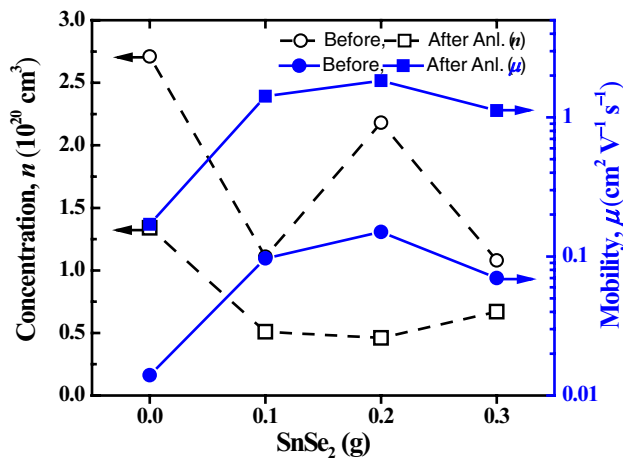


Fig. 5 The n and μ values of PEDOT:PSS + SnSe₂ polymer sheets as measured by the Hall effect method at room temperature.

after annealing depends on the transport mechanism of carrier hopping behaviors,²⁴ while the n values of the samples before and after annealing have displayed slight differences of around one order of magnitude (10^{20} cm³ (before) to 10^{19} cm³ (after)) within the significance. Moreover, the n and μ values could be affected to the electrical resistivity and Seebeck coefficient of the TE materials in theoretical calculations, as defined by Eqs. 3 and 4,³⁴ respectively:

$$\rho = \frac{1}{qn\mu}, \quad (3)$$

$$S = \frac{8\pi^2 k_B^2}{3eh^2} m_d^* T \left(\frac{\pi}{3n} \right)^{2/3} \quad (4)$$

where q , h , k_B , m_d^* , and e are carrier charges, the Planck constant, the Boltzmann constant, the density of states, effective mass of carrier, and elementary charge, respectively. Then, the electrical resistivity and Seebeck coefficient values or TE properties were measured and are discussed in the next section.

Figure 6 shows the TE properties composed of (a) ρ , (b) S , and (c) the PF of PEDOT:PSS + SnSe₂ polymer sheets at room temperature. The values of the samples before annealing were slightly decreased with a SnSe₂ concentration of 0.1 g and 0.2 g and then increased after a SnSe₂ concentration of 0.3 g. After the sample was annealed, the ρ values were slightly decreased and then trended stable with the SnSe₂ concentrations. The role of the RTA process induced the microstructure of SnSe₂ to decrease the ρ value by one order of magnitude (PEDOT:PSS + SnSe₂ (0.2 g); 1.91 mΩm (before) and 0.74 mΩm (after)) due to the mobility enhancement within the carrier hopping behaviors. While the S values for all samples indicated p -type

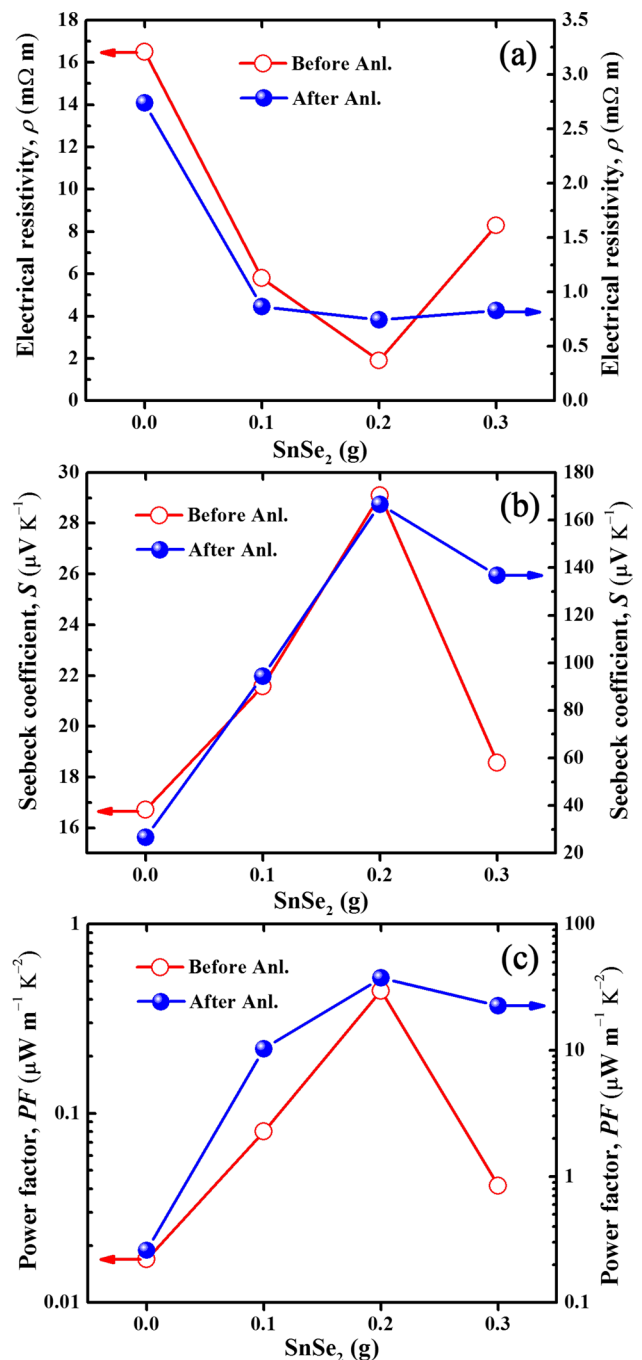


Fig. 6 (a) Electrical resistivity, (b) Seebeck coefficient, and (c) power factor of PEDOT:PSS, PEDOT:PSS + SnSe₂ (0.1 g), PEDOT:PSS + SnSe₂ (0.2 g), and PEDOT:PSS + SnSe₂ (0.3 g).

thermoelectricity, the S values of samples before annealing were slightly increased with a SnSe₂ concentrations of 0.1 g and 0.2 g and then increased after a SnSe₂ concentration of 0.3 g. After the sample was annealed, the S values suddenly increased with a SnSe₂ concentrations of 0.1 g and 0.2 g, and then they increased after a SnSe₂ concentration of 0.3 g. These results indicate that the increase in carrier mobility

Table I Comparison of PEDOT:PSS + SnSe₂ with other studies

Methods	Compositions	Temp. (K)	PF ($\mu\text{W m}^{-1} \text{K}^{-2}$)	References
Vacuum filtration	PEDOT:PSS/SnSe thin film	353	24.42	35
Vacuum filtration	PEDOT:PSS/SnSe nanosheet enhancing by SWCNT	380	109.9	36
Vacuum filtration	PEDOT:PSS/SnSe enhancing by MWCNTs	393	59.35	37
Cold-pressing & Vacuum filtration	PEDOT:PSS/SnSe- MWCNTs	393	166.5	38
Direct vacuum-filtration	PEDOT:PSS/MoSe ₂ nanosheets	300	48.6	39
Spin coating	PEDOT:PSS/Cu ₂ Se thin film	450	20.1	40
Wet chemical	PEDOT:PSS/SnSe ₂ polymer sheet	300	37.4	This work

MWCNTs multiwall carbon nanotubes, SWCNT single-walled carbon nanotube.

and the decrease in carrier concentration after annealing have been induced to enhance the Seebeck coefficient. As the same ρ values, the role of the RTA process has increased the S values by around one order of magnitude compared to before annealing (PEDOT:PSS + SnSe₂ (0.2 g); $29 \mu\text{V K}^{-1}$ (before) and $164 \mu\text{V K}^{-1}$ (after)). From the low ρ and high S values of PEDOT:PSS + SnSe₂ (0.2 g), the PF yielded a maximum of $37.4 \mu\text{W m}^{-1} \text{K}^{-2}$ after annealed at 220°C . In comparison, as listed in Table I, the TE properties of the PEDOT:PSS + SnSe₂ polymer sheets are comparable to the literature on doped PEDOT:PSS polymer sheets.^{35–40} The enhanced TE power factor is affected by changes in the crystallography and morphology within the increased crystallinity of SnSe₂ due to rapid thermal annealing.

Conclusions

We prepared organic/inorganic composite sheets based on PEDOT:PSS and SnSe₂ nanopowder. The microstructure of the sheets showed the general amorphous structure of PEDOT:PSS and then led the composite with the SnSe₂ phase after annealing. TE properties of the composite PEDOT:PSS + SnSe₂ sheets showed a p -type thermoelectricity at room temperature. The ρ and S were increased with the increase in the concentration of SnSe₂. The PEDOT:PSS + SnSe₂ (0.2 g) sample showed the maximum PF of $0.44 \mu\text{W m}^{-1} \text{K}^{-2}$ (before annealing) and $37.4 \mu\text{W m}^{-1} \text{K}^{-2}$ (after annealing) at 220°C , which indicated that the effect of RTA could enhanced the PF by around two orders of magnitude.

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Conflict of interest The authors declare no conflict of interest.

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